

BE/APh161: Physical Biology of the Cell

Homework 5

Due Date: Wednesday, February 18, 2026

“Thinking, analyzing, inventing are not anomalous acts; they are the normal respiration of the intelligence.” - Jorge Luis Borges

1. A First Cut at Exploratory Dynamics.

I provided you with three vignettes on what we are calling exploratory dynamics. As I argued there, we have had centuries of approaches to dynamics based on “update rules.” In this problem, you are going to recreate both the conceptual basis of that discussion and the math of the geometric distribution as one way to approach problems in exploratory dynamics.

(a) In this first part of the problem, give a one paragraph coherent summary of what I mean by update rules and how various examples from science that you are aware of either do or do not conform to my update rule paradigm.

(b) In this part of the problem, give a one paragraph coherent summary of what I mean by exploratory dynamics and provide several examples that illustrate the point.

(c) Now, let’s set up a generic approach to exploratory dynamics by considering that all the trajectories have the same basic structure. A series of wrong moves followed by a final right move. At each time step, the wrong and right moves are each chosen with probability f_w and f_r , respectively, subject to the condition that $f_w + f_r = 1$. A given exploratory trajectory is assembled as a collection of i unsuccessful steps before a final successful step. The probability of such a trajectory is given by

$$p_i = \underbrace{f_w f_w \cdots f_w}_i f_r = f_w^i f_r. \quad (1)$$

We can rewrite this more intuitively as

$$p_i = (1 - f_r)^i f_r, \quad (2)$$

a probability distribution known more colloquially as the geometric distribution. Your job in this part of the problem is to show that this distribution

is normalized and to compute the mean waiting time until the first success. Please make sure you explain how you do the math and also, what the math means.

2. The dynamics of ion channels.

(a) To exploit the rate-equation paradigm for the problem of ion channel dynamics, we begin with the simplest case by appealing to Figure 1. The idea is that we have a patch of membrane that is occupied by a total of N distinct two-state channels. For now, our goal is to write a rate equation that characterizes the time evolution of the probability of being in the open state and the way we will do this is to compute $p_{open}(t) = N_O(t)/N$ (where $N_O(t)$ is the number of open channels at time t) which will effectively determine that probability. We adopt a strategy in which time is discretized into steps of duration Δt and at each time step, the channel can undergo a transition from its current state or it can remain in the same state. The “reaction” of interest is of the form



where O signifies the open state, C signifies the closed state and k_+ and k_- are the rate constants that determine the probability of a change of state during a given time step. In particular, if the channel is closed, the probability it will make a transition to the open state in the time step Δt is $k_+\Delta t$. Similarly, if the channel is open, the probability it will make a transition to the closed state in the time step Δt is given by $k_-\Delta t$. Show that the change in the number of open channels in a given time step can be written as the update rule

$$\Delta N_O = \underbrace{-k_-\Delta t N_O}_{O \rightarrow C} + \underbrace{k_+\Delta t N_C}_{C \rightarrow O}. \quad (4)$$

Now show that if we divide both sides by Δt we find the rate of change in the number of open channels as

$$\frac{\Delta N_O}{\Delta t} = -k_- N_O + k_+ N_C, \quad (5)$$

which can be further simplified by dividing this equation by N and using $p_O = N_O/N$ and $p_C = N_C/N$. If we examine the limit as $\Delta t \rightarrow 0$, this

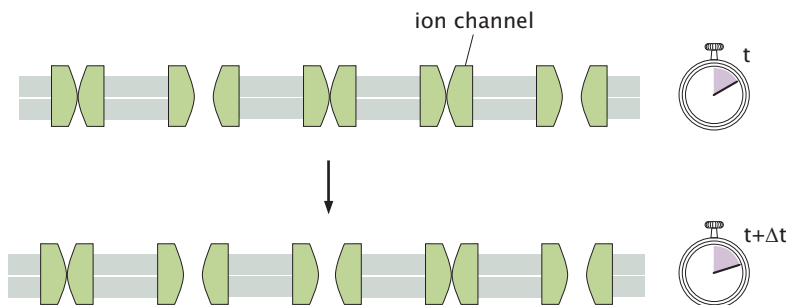


Figure 1: Channel gating kinetics. A patch of membrane with a collection of ion channels, some of which are open, some of which are closed. At every time step, channels can either switch their state or stay in the same state.

results in the differential equation

$$\frac{dp_O}{dt} = -k_-p_O + k_+p_C, \quad (6)$$

our first example of a chemical master equation, a dynamical description of the probabilities over time. If we now exploit the fact that $p_O + p_C = 1$, which amounts to the statement that the channels are either open or closed (that is, it is a two-state system) show that this may be rewritten as

$$\frac{dp_O}{dt} = -k_-p_O + k_+(1 - p_O). \quad (7)$$

This equation is more transparent if written in the form

$$\frac{dp_O}{dt} = -(k_- + k_+)p_O + k_+. \quad (8)$$

Solve that equation and imagine that at $t = 0$ all channels are in the closed state and give an expression for the probability of being open as a function of time. Make a graph that shows this transient dynamics and make sure you explain the steady state value of p_O in terms of the rate constants.

(b) In class, we considered an apparently completely different perspective on the opening probability of ion channels using the Boltzmann distribution. Repeat the steps done there by showing the “states and weights” for a two-state (i.e. closed and open) ion channel. Then, write an expression for p_O

in this statistical mechanical framework and plot the open probability as a function of the energy difference between the open and closed states. Explain what this open probability means.

(c) The problem with what we did in part (b) is that there is no stimulus that alters the difference in energy between the closed and the open state. Let's make a toy model of voltage gating of an ion channel by assuming that when the channel open, a charge moves in the direction perpendicular to the membrane with an energy $q\Delta V$. Now adjust your states and weights diagram of part (b) to include this voltage contribution to the energy of gating and compute a new version of the open probability p_O and then make a plot of p_O vs V . Explain the resulting plot. Please make sure that your numbers are plausible. By that I mean use the fact that in the absence of voltage, the closed state is stable. And, make sure that for typical transmembrane voltages that are characteristic of cells, that you favor the open state.

(d) Bob Austin of Princeton University liked to say “physics isn't worth a damn unless you put in some numbers.” In that spirit, in this part of the problem, we examine what these ion channels do. Figure 2(A) shows a single-channel recording of the current passing through a voltage-gated sodium channel. The data reveal that the channel transitions between open and closed states as shown in Figure 2(B). When in the open state, Na^+ ions can flow from one side of the membrane to the other, resulting in a current across the membrane. Given that ions have a typical diffusion constant of $1000 \mu\text{m}^2/\text{s}$, given the difference between the sodium intracellular and extracellular concentrations shown in Figure 2(C), and using a rough guess for the radius of an ion channel, use Fick's law to estimate the current that flows through the ion channel when in the open state. Recall that the charge of one monovalent ion is $1.6 \times 10^{-19} \text{ C}$ (Coulomb), and that $1 \text{ A} = 1 \text{ C/s}$ (Ampere = Coulomb/second). Compare your estimate to the current measured in Figure 2(A). Is pA the scale of currents that your estimates yield?

3. Setting up the fly body plan.

One of the most important ideas for how positional information arises in multicellular organisms is the idea of a morphogen gradient (another serious contender is a Turing pattern). In this problem we will use a steady-state solution to the reaction-diffusion equation for Bicoid to understand how the

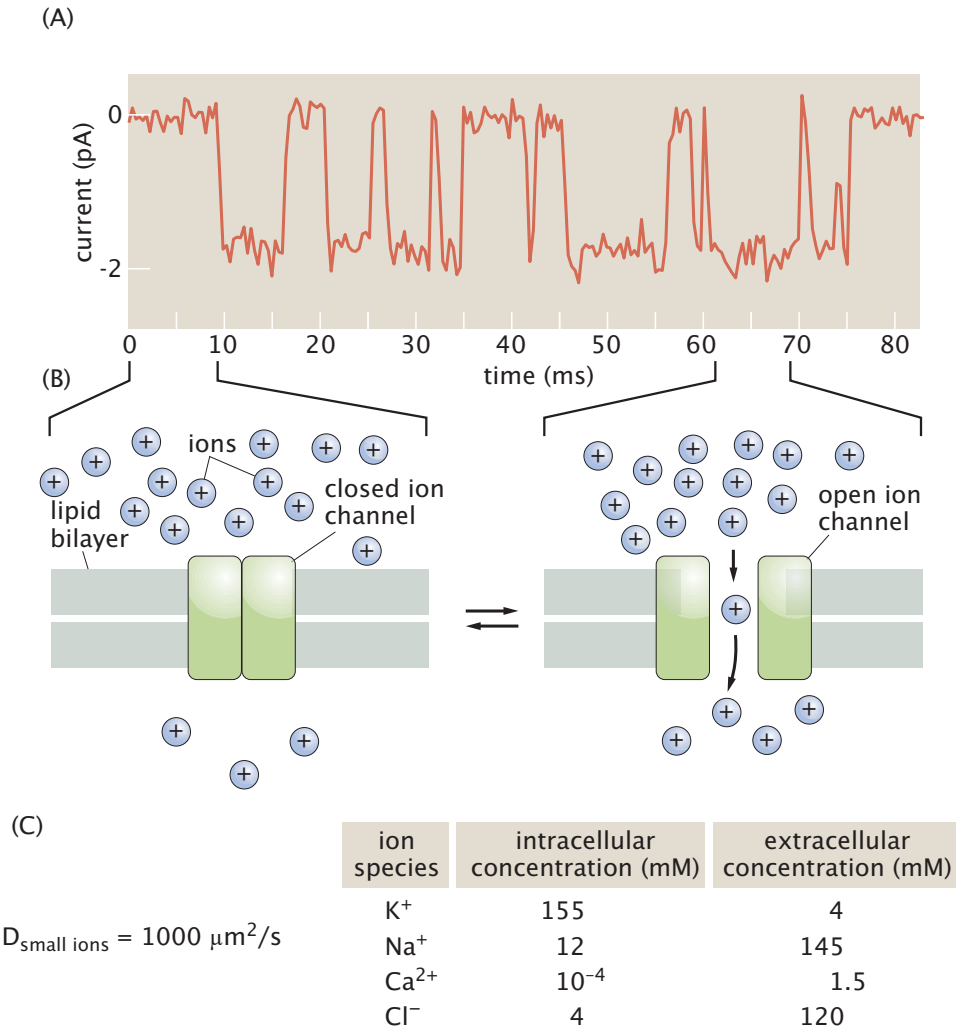


Figure 2: Electrical current flowing through an ion channel. (A) Current flowing through a single voltage-gated sodium channel. (B) The channel recording reveals transitions through an open and a closed state. (C) The concentration gradient of Na⁺ ions across the membrane can be used to estimate the current when the channel is open. (A, adapted from B. U. Keller et al., *J. Gen. Physiol.* 88:1, 1986; B, adapted from B. Hille, *Ion Channels of Excitable Membranes*. Sinauer Associates, 2001)

exponential profile shown in Figure 3 is set up. Stated simply, the development of the Bicoid gradient can be thought of as resulting from a competition

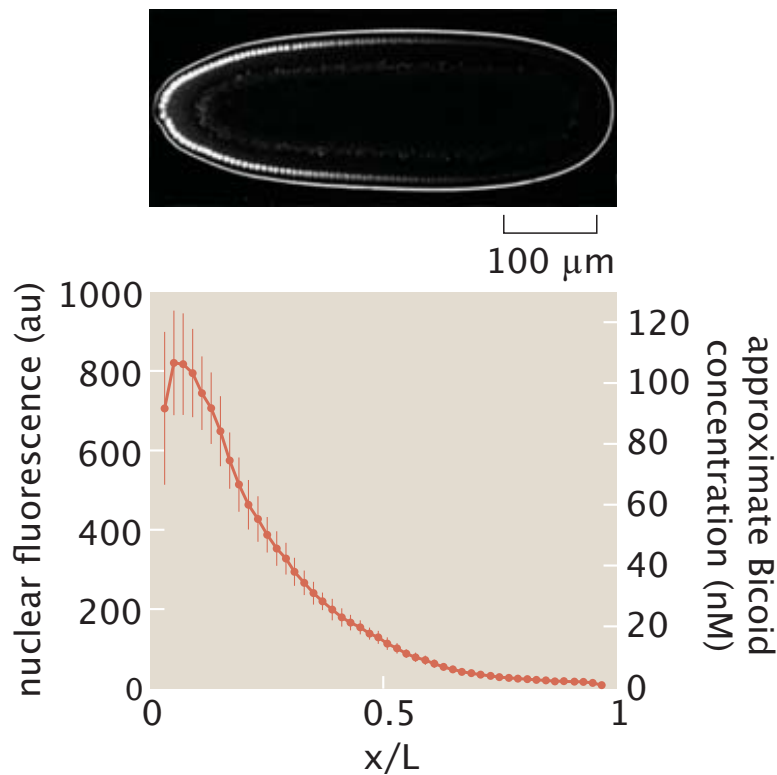


Figure 3: The Bicoid morphogen. The Bicoid activator is distributed in an exponential gradient. (Adapted from F. Liu *et al.*, Proc Natl Acad Sci USA 110:6724 2013.)

between the diffusion of Bicoid protein that is synthesized at the anterior end of the embryo (the mother deposits localized *bcd* mRNA there as shown in Figure 4) and the degradation of this protein while it is diffusing around.

(A) Give a brief description (a paragraph or less) of the Bicoid gradient in *Drosophila* and how it is relevant to fly development. Further, to get a feeling for the Bicoid gradient, redraw the Bicoid profile shown in Figure 3 in terms of the absolute number of Bicoid proteins per nucleus. You can make the drawing by hand or plot some approximate curve using Python. To make this estimate, you will need to use the information about nuclear sizes in nuclear cycle 14 provided in Figure 4C of Gregor2007a (provided on the course website).

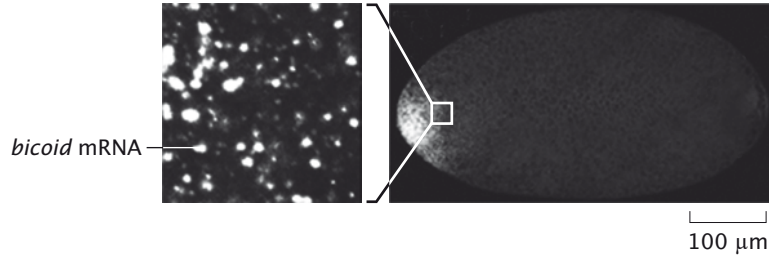


Figure 4: *bicoid* mRNA distribution. Using single molecule mRNA FISH, the localization of individual *bicoid* mRNA molecules at the anterior end of the embryo can be revealed. (Adapted from Petkova et al. (2014), *Current Biology* 24:1283.)

(B) Make a derivation of the reaction-diffusion equation and use it to justify the form

$$\frac{\partial Bcd(x, t)}{\partial t} = D \frac{\partial^2 Bcd(x, t)}{\partial x^2} - \frac{Bcd(x, t)}{\tau}. \quad (9)$$

Make sure you explain carefully where all of these terms come from. To do so, begin the usual way by considering a one-dimensional concentration profile and by finding the rate of change of number of Bicoid molecules in the box at position x by considering the flux into ($J_m(x - \Delta x/2)$) and out of ($J_m(x + \Delta x/2)$) the box using arguments like those made in class. However, you need to generalize that treatment by accounting for the fact that a Bicoid molecule has the probability $r\Delta t$ of degrading in time interval Δt , where $r \approx 1/\tau$, where τ is the degradation time.

(C) Now solve this equation in steady-state by finding the general solution subject to the boundary condition that $J(0, t) = j_0$ and $J(L, t) = 0$. Make sure you explain what these boundary conditions mean relative to the biology of the problem. Suggest approximations that can be made to simplify the result, specifically, can you exploit the fact that the embryo is much larger than the decay length to simplify the solution?

(D) Describe the observed concentration profile of Bicoid along the anterior-posterior axis of the fly mathematically. What is the functional form? Experimentally, Thomas Gregor has found that the Bcd profile is an exponential of the form $Bcd(x) = Bcd_0 e^{-x/\lambda}$ where x is the position along the embryo,

Bcd_0 is the Bicoid concentration at $x = 0$ and λ is the decay constant of the gradient. Does that experimental profile jibe with your solution?

(E) The paper by Drocco *et al.* uses a photoactivatable fluorescent protein to measure the lifetime of the Bicoid protein. Read the paper (available on the course website) and explain the technique in one paragraph. You might find it useful to draw a schematic plot such as shown in Figure 1f of the paper.

(F) What is the value of the decay constant λ for the gradient shown in Figure 3? To estimate this magnitude, you can just fit “by eye” by plotting your solution for different values of Bcd_0 and λ . Now, compare the measured λ value with that you can predict by plugging in realistic values of D , τ into your solution. To make this possible, read the papers by Abu-Arish *et al.* and Drocco *et al.*, provided on the course website.

(G) One of the most important and interesting ideas to come out of the idea of positional information contained in morphogen gradients was the so-called French flag model which we will explore here. This model posits that the Bicoid concentration dictates the position of the cephalic furrow. As seen in Figure 5, the idea of the model is that boundaries in the embryo are determined by threshold values of the morphogen. The idea of the model is that if the gene dosage gets changed, as seen in the mutant profile, the boundary will still occur at the same value of the morphogen. That hypothesis is enough to determine the shift in boundary position with gene dosage.

To test this model, we will analyze several experiments (Nusslein-Vohldhard and Driever and Liu *et al.*) where they measured cephalic furrow position as a function of different dosages of the *bicoid* gene in embryos. As seen above, an exponential gradient of Bicoid is described by

$$Bcd(x, \lambda, \alpha, Bcd_0) = Bcd_0 \alpha e^{-x/\lambda}, \quad (10)$$

where x is the position along the embryo, Bcd_0 is the Bicoid concentration at $x = 0$, λ is the decay constant of the gradient and α is the Bicoid dosage, with $\alpha = 1$ corresponding to the wild-type. Work out a model for the position of the cephalic furrow x_{new} as a function of the gene dosage α , the morphogen gradient decay length λ and the position of the wild-type cephalic furrow, x_{CF} .

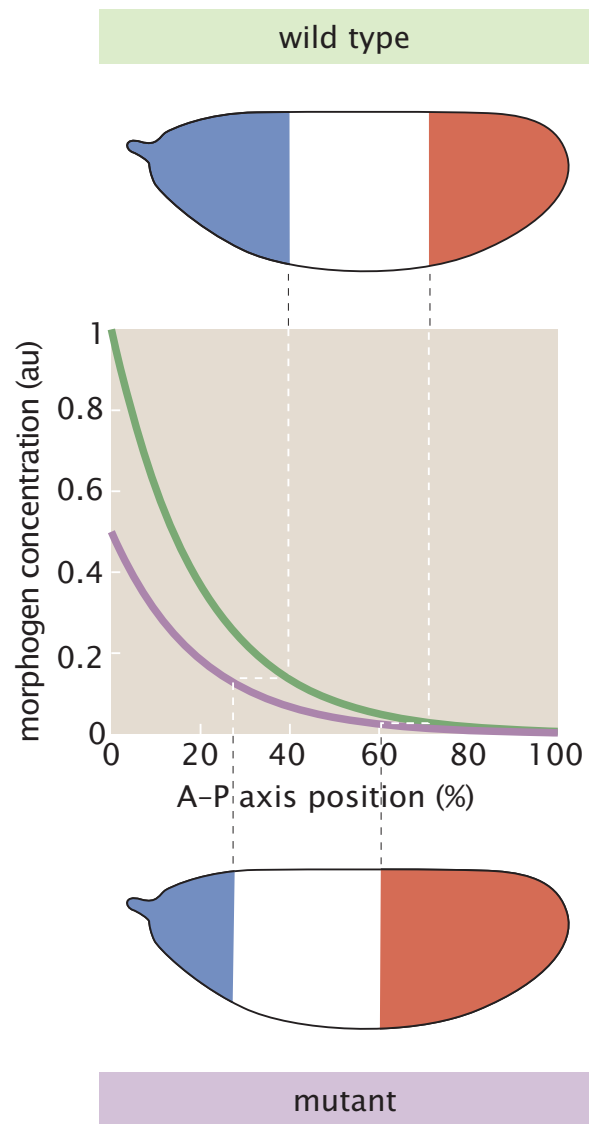


Figure 5: Concept of the French flag model. The green profile shows the wild-type morphogen concentration and the purple profile shows that of a mutant with half the wild-type gene dosage.

(H) Note that, given a measured $x_{CF} \approx 32\%$ of the embryo length, your model has no free parameters. Compare the prediction from your model with the data for x_{new} vs. α obtained by Nusslein-Vohlhard, and by Driever and Liu *et al.*. Comment on how well your prediction matches the data that is provided with the homework. What could be going on?

4. Diffusive speed limits: It's not just a good idea, it's the law

In order for a chemical reaction to take place, the reactants must be at the same place at the same time. A very interesting calculation explores the way in which diffusion can control the on rate for reactions. Imagine some reaction in which A and B come together to form the complex AB . To simplify the problem, we are going to imagine B as a sphere of radius a that is fixed at the origin of our coordinate system. Further, we are going to imagine that very far away the concentration of A is held at c_0 . What I really mean by this is that $\lim_{r \rightarrow \infty} c(r) = c_0$, where $c(r)$ is the concentration of reactant A as a function of distance from the origin. Our goal is to compute the so-called “diffusion-limited on rate” for the reaction. We begin by working out the steady-state solution to the diffusion equation with the boundary condition that $c(a) = 0$, which corresponds to the physical statement that the sphere is a “perfect absorber”. What this really means is that every time a molecule of A arrives at the sphere, the reaction occurs. (Note that this tells us that the diffusion-limited on rate is the fastest that a reaction could occur. It could be true that after the molecule arrives, it has to wait for some favorable orientation to occur, for example, which would make the rate of the reaction even slower).

(a) Solve the diffusion equation

$$\frac{\partial c(\mathbf{r}, t)}{\partial t} = D \nabla^2 c(\mathbf{r}, t) \quad (11)$$

in steady state and find the concentration profile $c(r)$ as a function of c_0 and a . Explain why we can write the concentration only as a function of the scalar r as opposed to the vector $\mathbf{r}$.

(b) Use that result to compute the diffusive flux $J(a)$ at the surface of the sphere. Here you need to invoke Fick's law relating flux and concentration,

but acknowledging that you are working in spherical coordinates.

(c) Use the result of part (b) to write an equation for dn/dt , the rate at which A molecules arrive at the sphere and thus the rate of production of AB . The function $n(t)$ simply tells me how many molecules have arrived at the “perfect absorber” during the time between $t = 0$ and the time t .

(d) Now, use the result of part (c) to write an equation of the form

$$\frac{dn}{dt} = k_{on}c_0, \quad (12)$$

and hence write an expression for k_{on} . This is the so-called Smoluchowski rate.

(e) Find a numerical value for this diffusion limited on rate, k_{on} . Justify the units it has and provide an actual numerical value by estimating the relevant parameters that determine k_{on} .

(f) In a few sentences, explain how this picture allows us to think about the on-rates for chemical reactions.